Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/22/2006
PI Name:	Schiffer, Peter Ph.D.		
Project Title:	Correlation Effects in Dense Wet and Dry Granular Media		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (12/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNC04GA33G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This grant is supporting research into the fundamental properties of granular media. There is a particular focus on the properties of dense granular media and the development of correlations therein. A better understanding of the development and evolution of correlations in dense granular media. Granular media are used in a wide range of industrial processes, and understanding their properties is important to improving these processes. NASA missions (such as setting up lunar or martian bases) will also require extensive handling of granular materials, and therefore the understanding of these materials will be important for the design and implementation of those missions. Finally, the understanding of granular materials is of great fundamental scientific importance.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	More efficient industrial processes and better understanding of fundamental science always improves life on earth. This research should yield advances in both.
Task Progress:	We have initiated three new experimental efforts which are all yielding interesting results. Penetration Experiments: The first experiment is analogous to the completed penetration experiment described previously, except that we are measuring the force required to push a solid object upwards through a granular pile from the bottom. This will reveal how local jamming is affected by the presence of a free boundary in the material. Preliminary data show interesting effects which were not anticipated, but which need to be rigorously verified in the upcoming months. This line of research has relevance to the Moon/Mars missions in that parts of those missions will involve pushing probes through granular layers on the surfaces of the Moon and Mars. We are trying to understand the fundamental physics of pushing probes through granular materials. Thermal Cycling Experiments: The second experiment is a study of thermal cycling effects in granular materials. The process of thermal cycling in granular media is different from that in homogeneous solids. In homogeneous solids, raising or lowering the temperature leads to a homogeneous change in the lattice constant of the material and thus to a reversible expansion or contraction of the material. If one returns to the same temperature, one expects the material to be unchanged. By contrast, in a granular material, the individual solid grains can move relative to each other and they are typically in some random configuration which is only incrementally separated from other configurations. The small changes in grain size which accompany heating or cooling the grains, combined with gravitational forces on the grains, can thus cause irreversible changes in the grain configurations. This can lead to changes in packing fraction, breaking of grains, and possibly other more exotic effects such as segregation of different grain sizes and the onset of avalanches and container failure. We are investigating these effects in model granular systems. This line of research has relevance to the Moon/Mars missions in that there are extremely large daily thermal cycles in both the lunar and Martian environments. Understanding the consequences for the handling of granular materials will be crucial to avoid container rupture and other undesired results. Granular Flow From Vibration Experiments: The third experiment is a study of granular flow through an orifice in the bottom of a vibrating container. The goal of this experiment is to eventually study the development of correlations in wet granular materials in which the liquid content results in intergrain cohesion. We have begun this study by examining the flux of dry grains as a function of the frequency and amplitude of the vibrations. Even the dry case is extraordinarily rich in physical phenomena, and these must be understood before we can consider the wet case. We have found extraordinarily rich behavior which can be well modeled with a hydrodynamic theory showing that the velocity amplitude of the vibration is the controlling parameter of the flow. This line of research has relevance to the Moon/Mars missions in that the flux of grains from a container is an important process for many applications, so understanding the physics of the process will be important for reduced gravity operations.
Bibliography Type:	Description: (Last Updated: 11/15/2007)
Articles in Peer-reviewed Journals	Chen K, Stone MB, Barry R, Lohr M, McConville W, Klein K, Sheu BL, Morss AJ, Scheidemantel T, Schiffer P. "Flux through a hole from a shaken granular medium." Phys Rev E Stat Nonlin Soft Matter Phys. 2006 Jul;74(1 Pt 1):011306. Epub 2006 Jul 25. PMID: 16907088 , Jul-2006
Articles in Peer-reviewed Journals	Chen K, Cole J, Conger C, Draskovic J, Lohr M, Klein K, Scheidemantel T, Schiffer P. "Granular materials: packing grains by thermal cycling." Nature. 2006 Jul 20;442(7100):257. PMID: 16855580 , Jul-2006
Presentation	Schiffer, P. "Local Jamming: Drag and Penetration Through Granular Media " Meeting of the New York Section of the American Physical Society, Rochester, NY, April 16, 2005. Apr-2005
Presentation	Schiffer, P. "Beach Physics" Condensed Matter Physics Seminar, Physics Dept., University College London, United Kingdom, January 24, 2005. Jan-2005
Presentation	Schiffer. P. "Drag and Penetration through Granular Materials" Conference on Granular Physics, Kavli Institute for Theoretical Physics, Santa Barbara, CA, June 23, 2005. Jun-2005
Presentation	Schiffer. P. "Drag and Penetration through Granular Materials" Conference on Granular Physics, Kavli Institute for Theoretical Physics, Santa Barbara, CA, June 23, 2005. Jun-2005